MSTool-AI

Team Operating Guide

IEC 62304 Class C Compliance

Document ID	Version	Date	Audience
GUIDE-001	**1.0**	**April 12, 2026**	**ALL Team Members**

! CRITICAL SAFETY NOTICE

MSTool-AI is classified as IEC 62304 Software Safety Class C — the HIGHEST safety class.
A software failure could contribute to DEATH or SERIOUS INJURY.
Every person working on this software MUST follow these procedures.
A government audit will verify compliance for EVERY change.

? The Auditor Will...

Pick random Git commits and trace them to requirements, risk analysis, and tests
Pick random requirements and trace them to code and verification evidence
Ask to see signed review records, test results, and problem reports
Check that templates (PDFs in docs/iec62304/templates/) are being used

Contents

1 FOR DEVELOPERS — Daily Workflow

1 Before Starting Any Work

- [ ] Check GitHub Issues for assigned task
- [ ] Identify which requirement(s) the task implements
  - → Open `docs/iec62304/02_Software_Requirements_Specification.md`
  - → Find the REQ-ID (e.g., REQ-FUNC-040)
  - → **If no requirement exists, STOP and create one first**
- [ ] Check if the task touches a Class C module
  - → If YES: read the risk analysis in RMF-001
  - → If YES: your PR will need extra scrutiny (2 reviewers)
- [ ] Create a feature branch from main

```
git checkout -b feature/REQ-FUNC-040-description
```

2 While Writing Code

Follow coding standards:

- → TypeScript: ESLint rules + TypeScript strict mode
- → Python: PEP 8 + type annotations

For Class C modules:

- [ ] Add input validation for ALL external inputs
- [ ] Handle ALL error cases (no silent failures)
- [ ] Add unit tests for the change
- [ ] Document safety-related behavior in code comments

Commit message format:

```
git commit -m "feat(REQ-FUNC-040): Implement volume computation

  Implements voxel counting with normative percentile comparison.
  Risk control RC-004 (volumetry displays percentile ranges).
  Unit test: UT-VOL-001"
```

3 Submitting a Pull Request

PR Title: [REQ-ID] Brief description

Example: `[REQ-FUNC-040] Brain volumetry computation`

PR Description MUST include:

- [ ] What: description of the change
- [ ] Why: which requirement it implements
- [ ] Risk: which hazards are affected (if any)
- [ ] Tests: which tests verify the change
- [ ] SOUP: any new dependencies added?

> ✓ **CI Pipeline — ALL must be GREEN before requesting review**
> TypeScript check (npx tsc --noEmit)
> Frontend build (npm run build)
> Backend tests (python -m pytest)
> SOUP vulnerability scan (npm audit + pip-audit)
> Python syntax check (ast.parse all .py files)

- [ ] Request review from at least 1 team member
- [ ] For Class C changes: request review from 2 team members

4 Code Review (as Reviewer)

> **T Use Template: TPL-03 Code Review Checklist**
> Location: docs/iec62304/templates/TPL-03_Code_Review_Checklist.pdf
> This filled form is AUDIT EVIDENCE — it must be saved!

General Checks:

- [ ] Code implements the detailed design (DD-001)
- [ ] Coding standards followed
- [ ] No commented-out code without issue link

Safety Checks (Class B/C):

- [ ] Input validation for all external inputs
- [ ] Error handling covers all failure modes
- [ ] No unhandled exceptions
- [ ] Risk controls correctly implemented

Class C Specific:

- [ ] No unintended functionality
- [ ] Robustness with invalid inputs
- [ ] SOUP APIs used correctly

Save the filled checklist:

```
Filename: CR-YYYY-NNN_[module_name].pdf
Store in: docs/iec62304/records/code_reviews/
```

5 After Merge and Deploy

For backend changes:

```
gcloud builds submit --config=cloudbuild.yaml \
  --substitutions=COMMIT_SHA=$(git rev-parse --short HEAD)

# Wait for build SUCCESS, then:
bash test_endpoints.sh    # ALL 9 checks must PASS
```

For frontend changes:

```
cd frontend && npm run build
npx firebase deploy --only hosting
# Verify the app loads correctly in browser
```

2 FOR QA / TESTERS — Testing Workflow

Testing a Requirement

T Use Template: TPL-06 Test Execution Report
Location: docs/iec62304/templates/TPL-06_Test_Execution_Report.pdf

- [ ] Find the requirement in SRS-001
- [ ] Fill in ALL fields: Test ID, Requirement, Software Version, Environment, Tester, Date
- [ ] Write clear test steps (preconditions, step-by-step, expected result)
- [ ] Execute test and document actual result + PASS/FAIL
- [ ] Save: ST-FUNC-XXX_[date].pdf in docs/iec62304/records/test_results/

Testing a Risk Control

T Use Template: TPL-04 Risk Control Verification
Location: docs/iec62304/templates/TPL-04_Risk_Control_Verification.pdf
There are 22 risk controls (RC-001 through RC-022) — ALL must be verified.

- [ ] Find risk control in RMF-001 (docs/iec62304/03_Risk_Management_File.md, Section 5)
- [ ] Verify the control is implemented in the code
- [ ] Test that it works correctly
- [ ] Document verification method and result
- [ ] Sign and date
- [ ] Save: RCV-RC-XXX_[date].pdf in docs/iec62304/records/risk_verification/

3 FOR PROJECT LEAD — Management Workflow

Release Process

- [ ] Fill the Pre-Release Checklist (TPL-02_Release_Checklist.pdf)
- [ ] All 11 checks must be marked YES
- [ ] Document any known anomalies with risk assessment
- [ ] Sign the approval section

Quality Gates: Use TPL-10 for each gate:

G1 (Requirements) → G2 (Design) → G3 (Implementation) → G4 (Integration) → G5 (Release)

Monthly Tasks

M SOUP Vulnerability Review — MONTHLY (Mandatory)

Use TPL-07_SOUP_Vulnerability_Review.pdf
Run: npm audit (frontend) + pip-audit (backend)
Check NVD for 7 Class C SOUP items
Document findings and actions, sign and date
Store in: docs/iec62304/records/soup_reviews/

Q Risk Management File Review — QUARTERLY

Review RMF-001 for new hazards
Check if any changes introduced new risks
Update if needed, document review in RMF-001 Section 7

When a Bug is Found

- [ ] Create GitHub Issue using problem report format
- [ ] Use TPL-01_Problem_Report.pdf for formal documentation

! CRITICAL: Always Assess Safety Impact

Does this affect a Class C module?
Could this contribute to a hazardous situation?
Does this require regulatory notification (EU MDR Article 87)?
If safety-impacted: Use TPL-08, notify authority within 24 hours if serious

Document Approval

Use **TPL-11_Document_Approval.pdf** — all 15 IEC 62304 documents need formal approval signatures.

Each document needs: Reviewer name + Approver name + Date + Signature.

Store signed original in: `docs/iec62304/records/approvals/`

4 FOR CLINICAL ADVISOR — Clinical Workflow

Risk Analysis Review

- [ ] Review RMF-001: Are all 14 hazardous situations clinically accurate?
- [ ] Are severity ratings appropriate?
- [ ] Are risk control measures clinically adequate?
- [ ] Is the benefit-risk analysis justified?
- [ ] Sign off using TPL-05_Design_Review_Record.pdf

Requirements Review

- [ ] Review safety requirements (REQ-SAFE-001 through REQ-SAFE-020)
- [ ] Are disclaimers clinically appropriate?
- [ ] Is the "assistive tool" positioning clear?
- [ ] Are confidence thresholds clinically meaningful?
- [ ] Are AI performance thresholds appropriate?
- [ ] Is the de-identification approach adequate for HIPAA?

5 Record Keeping — Where to Store Everything

Folder Structure

```
docs/iec62304/records/
■■■ code_reviews/           ← Filled TPL-03 for each PR
■   ■■■ CR-2026-001_volumetry_service.pdf
■   ■■■ ...
■■■ test_results/           ← Filled TPL-06 for each test
■   ■■■ ST-FUNC-001_load_nifti.pdf
■   ■■■ ...
■■■ risk_verification/      ← Filled TPL-04 for each risk control
■   ■■■ RCV-RC-001_ai_disclaimer.pdf
■   ■■■ ...
■■■ soup_reviews/           ← Filled TPL-07 monthly
■   ■■■ SOUP-2026-04.pdf
■   ■■■ ...
■■■ releases/               ← Filled TPL-02 per release
■■■ design_reviews/         ← Filled TPL-05 per review
■■■ incidents/              ← Filled TPL-08 if needed
■■■ changes/                ← Filled TPL-09 per change
■■■ gate_approvals/         ← Filled TPL-10 per gate
■■■ approvals/              ← Filled TPL-11 document sign-off
```

Naming Convention

```
[Template Type]-[ID]_[Description]_[Date].pdf
```

Example Filename	Description
CR-2026-001_brain_volumetry_service_2026-04-15.pdf	Code review record
ST-FUNC-040_compute_volumes_2026-04-16.pdf	Test execution report
RCV-RC-004_volumetry_percentile_display_2026-04-17.pdf	Risk control verification
SOUP-2026-04_monthly_review.pdf	Monthly SOUP vulnerability review
REL-v2.0.0_release_checklist_2026-04-20.pdf	Release checklist

6 Quick Reference — What Template to Use When

Situation	Template	File
I'm reviewing someone's code	Code Review Checklist	TPL-03
I'm deploying a new version	Release Checklist	TPL-02
I found a bug	Problem Report	TPL-01
I'm testing a requirement	Test Execution Report	TPL-06
I'm verifying a risk control	Risk Control Verification	TPL-04
I'm reviewing architecture/design	Design Review Record	TPL-05
Monthly SOUP vulnerability check	SOUP Vulnerability Review	TPL-07
A serious safety incident occurred	Serious Incident Report	TPL-08
Making a significant code change	Change Control Record	TPL-09
Approving a development phase gate	Quality Gate Approval	TPL-10

Situation	Template	File
Formally approving all documents	Document Approval Record	TPL-11

7 Audit Preparation Checklist

When the audit is announced, verify ALL of the following:

- [ ] All 15 IEC 62304 documents exist and are current version
- [ ] All 11 PDF templates have been used (filled records exist)
- [ ] Document Approval Record (TPL-11) is signed by authority
- [ ] At least 1 filled Code Review Checklist per Class C module change
- [ ] At least 1 filled Test Execution Report per "Must" requirement
- [ ] All 22 Risk Control Verification records filled and signed
- [ ] Monthly SOUP Vulnerability Reviews exist for the past 6 months
- [ ] Release Checklists exist for each deployed version
- [ ] Git history shows PR reviews on all changes to main branch
- [ ] CI pipeline results are available (GitHub Actions artifacts)
- [ ] test_endpoints.sh results available for pre/post deploy
- [ ] No critical/major open bugs without safety assessment

8 Common Mistakes to Avoid

Mistake	Why It's a Problem	What To Do Instead
Pushing directly to main	No review evidence = audit failure	Always use pull requests
"Fix typo" commit without PR	Uncontrolled change = finding	Every change needs a PR
No commit message references	Auditor can't trace changes	Include REQ-ID or issue #
Skipping test_endpoints.sh	Can't prove deploy verified	Run BEFORE and AFTER deploy
Not filling PDF templates	No audit evidence exists	Fill templates, save as records
Ignoring SOUP vulnerabilities	Cybersecurity non-compliance	Monthly review is MANDATORY
Changing Class C without review	IEC 62304 5.5.4 violation	ALWAYS get code review
Not assessing safety of bugs	ISO 14971 violation	Every bug must assess safety

9 Contact and Escalation

Issue	Contact	Escalation
Code review question	Assigned reviewer	Project Lead
Safety concern about a change	Project Lead	Clinical Advisor
Serious incident suspected	Project Lead + Clinical Advisor	Regulatory Authority (BfArM) within 24h
SOUP vulnerability (critical)	Developer	Project Lead for risk assessment
Audit preparation question	Project Lead	Regulatory Affairs

References

All IEC 62304 documentation: `docs/iec62304/`

All fillable templates: `docs/iec62304/templates/`

All filled records: `docs/iec62304/records/`

Document	ID	Purpose
Master Compliance Document	00	Overall compliance status
Software Requirements Spec	SRS-001	All 91 requirements
Risk Management File	RMF-001	14 hazards, 22 risk controls
Detailed Design Specification	DD-001	Class C unit designs
Traceability Matrix	TM-001	Requirement ↔ test mapping

This guide must be read by every team member before contributing to MSTool-AI. Acknowledgment of reading this guide should be documented.

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